To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

In [2]:
```
%pip install nltk
```

```
Requirement already satisfied: nltk in /opt/miniconda3/lib/python3.14/site-p
ackages (3.10.3)
Requirement already satisfied: defusedxml in /opt/miniconda3/lib/python3.14/
site-packages (from nltk) (0.7.1)
Requirement already satisfied: click in /opt/miniconda3/lib/python3.14/site-
packages (from nltk) (8.4.1)
Requirement already satisfied: joblib in /opt/miniconda3/lib/python3.14/site
-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /opt/miniconda3/lib/python
3.14/site-packages (from nltk) (2026.9.29)
Requirement already satisfied: tqdm in /opt/miniconda3/lib/python3.14/site-p
ackages (from nltk) (4.68.2)
Note: you may need to restart the kernel to use updated packages.
```

In [1]:
```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
```

In [3]:
```
import nltk
```

In [4]:
```
nltk.download('gutenberg')
```

```
[nltk_data] Downloading package gutenberg to
[nltk_data]     /Users/harshitha/nltk_data...
[nltk_data]   Unzipping corpora/gutenberg.zip.
```

Out[4]:
```
True
```

In [5]:
```
nltk.download('punkt')
```

```
[nltk_data] Downloading package punkt to /Users/harshitha/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
```

Out[5]:
```
True
```

In [7]:
```
nltk.download('punkt_tab')
```

```
[nltk_data] Downloading package punkt_tab to
[nltk_data]     /Users/harshitha/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
```

Out[7]:
```
True
```

In [8]:
```
sample = gutenberg.raw("austen-emma.txt")
token = word_tokenize(sample)
```

In [10]:
```
wlist = []
for i in range(50):
 wlist.append(token[i])
# Calculate the frequency of each word in the list
 wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
 print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

```
Pairs
[('[', 1)]
Pairs
[('[', 1), ('Emma', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 1), ('CHAPTER', 1)]
Pairs
[('[', 1), ('Emma', 1), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 1), ('handsome', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 2), ('handsome', 1), (',', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 2), ('handsome', 1), (',', 2), ('clever', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 3), ('handsome', 1), (',', 3), ('clever', 1), (',',
3)]
Pairs
```

[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 3), ('handsome', 1), (',', 3), ('clever', 1), (',',
3), ('and', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 3), ('handsome', 1), (',', 3), ('clever', 1), (',',
3), ('and', 1), ('rich', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 1), ('rich', 1), (',', 4)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 1), ('rich', 1), (',', 4), ('with', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 1), ('rich', 1), (',', 4), ('with', 1), ('a', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 1), ('rich', 1), (',', 4), ('with', 1), ('a', 1), ('comfortabl
e', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 1), ('rich', 1), (',', 4), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 2), ('rich', 1), (',', 4), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 2), ('rich', 1), (',', 4), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 2), ('happy', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 4), ('handsome', 1), (',', 4), ('clever', 1), (',',
4), ('and', 2), ('rich', 1), (',', 4), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),

(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 1), ('the', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 1), ('the', 1), ('best', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),

(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 1), ('the', 1), ('best', 1), ('blessings', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2), ('existence', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 2), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 2), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('and', 3)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('and', 3), ('had', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1), ('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an

```
d', 3), ('had', 1), ('lived', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1), ('year
s', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 1),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1), ('year
s', 1), ('in', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 2),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1), ('year
s', 1), ('in', 1), ('the', 2)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 1), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 2),
```

```
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1), ('year
s', 1), ('in', 1), ('the', 2), ('world', 1)]
Pairs
[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816', 1),
(']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma', 2),
('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever', 1), (',',
5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1), ('comfortabl
e', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposition', 1), (',', 5),
('seemed', 1), ('to', 1), ('unite', 1), ('some', 1), ('of', 2), ('the', 2),
('best', 1), ('blessings', 1), ('of', 2), ('existence', 1), (';', 1), ('an
d', 3), ('had', 1), ('lived', 1), ('nearly', 1), ('twenty-one', 1), ('year
s', 1), ('in', 1), ('the', 2), ('world', 1), ('with', 2)]
```

In [ ]: